Department of Economics

MSc Extended Essay Cover Sheet

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TITLE	**Covid-19 & Remote Work: Examining the Accessibility and Impacts of Working from Home in the United States**

WORD COUNT	5	5	7	4

Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text). Examiners are recommended to penalise excessively long essays (+ 10% of word count) by adjusting the marks by 6000/x, where x is the word count. Pages are to be numbered.*

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COVID-19 & Remote Work: Examining the Accessibility and Impacts of Working from Home in the United States

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Abstract

In this paper, we will explore how wages and income vary across industries and illustrate this by using the finance and manufacturing industries as proxies for industries that have high ability to work from home and low ability to work from home. Survey responses of the IPUMS CPS, a representative sample of the United States population, showed that 24.7% of respondents worked from home in 2020 due to COVID-19. Among them, those that were able to work from home tended to have higher levels of education, earned more, were younger and female. Differences within industries are examined by comparing employees in the finance industry, who on average earn higher wages and are more educated, with those employed in finance. This report examines those who are employed in the finance industry were 35.4% more likely to work from home than those employed in manufacturing. Using these two assumptions, we can estimate a 12% wage premium from being able to work remotely. This is attributed to higher paying industries being able to work from home more as well as high ability workers within industries have more feasibility of remote work.

Keywords COVID-19, working from home, Mincer Model, Logistic Model
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1. Introduction

In the beginning of 2020, the novel Coronavirus was quickly spreading around the globe. World leaders responded by increasing security measures to further prevent the spread of the virus and protect their citizens. As lockdown orders were put in place, businesses began to worry about how to retain their employees and keep their companies afloat. One way of doing this was to have employees switch to remote work. However, not every occupation is suitable for remote work.

This sudden transformation of the labour market has sparked the interest of economic researchers around the world. Most researchers agree that remote work helps shield people from the economic detriments of the pandemic. This is largely due to the ability to keep one's job in the face of societal shutdowns as well as minimize the risk of exposure to the deadly virus. However, ability to work from home varies greatly and the long-term effects and persistence of this change in work culture is still up for debate.

This paper aims to explore the disparities in feasibility of remote work and better understand the consequences of being able to do so. By focusing on the labor force in the United States, this paper utilizes cross sectional data to identify those that are more likely to work from home based on worker demographics and industry of employment. The results show that the ability to work remotely is enjoyed by people who are better educated and highly paid. Within industries, high ability/paid employees are more likely to work remotely.

The remainder of the paper is structured as follows. In Section 2, recent economic literature is reviewed and discussed. Section 3 describes the data and examines accessibility of remote work for the IPUMS CPS respondents. Following this, Section 4 details the empirical strategy employed throughout the paper to examine the benefits of working from home. The main results of the analysis are presented in Section 5 while a series of robustness checks are conducted in Section 6. Section 7

concludes the research, with references and appendices appearing in Sections 8 and 9, respectively.

2. Literature Review

The COVID-19 pandemic has dominated subjects of discussion for the past two years, involving topics including employment, labour market losses, learning losses, social mobility, and scarring. While many businesses were forced to close their offices due to the pandemic induced shutdowns, others shifted to a remote labor force. By analyzing survey results from the biweekly Real-Time Population Survey (RPS), Bick and his co-authors estimate that 35.2% of the United States labor force worked from home in May of 2020, compared to 8.2% in the months before the pandemic (Bick et al., 2020). This significant and sudden augmentation of the remote workforce has become an area of intrigue among economic researchers due to the unprecedented nature of the labor shift.

However, remote work is not equally dispersed. The consensus among economic researchers is that working remotely is more likely to be an option for workers that have higher levels of education as well as higher wages and income levels. Using data from the Occupational Information Network, Mongey, Pilossoph, and Weinberg find that those without the ability to work from home are 40 percentage points less likely to have a college degree and 23 percentage points more likely to earn less than the median wage (Mongey et al., 2020). This disparity is likely because highly educated and highly paid workers are more likely to be employed in industries that have greater feasibility of switching to remote work.

These inequalities in the capability to switch to remote work during times of uncertainty such as the COVID-19 pandemic are significant because the alternative to working from home during a lockdown is job loss. Adams-Prassl and his co-authors studied labor market impacts of COVID-19 using survey data, comparing employment trends in the United States, United

Kingdom, and Germany. In the United States, 20% of laborers lost their job in April 2020 (Adams-Prassl et al., 2020). While this number may be staggering, 40% of laborers that were not able to work from home lost their job. The authors' findings are consistent with that of Mongey et al. who constructed a work from home measure across occupations, designating certain jobs as "High Work from Home" and "Low Work from Home" and estimated that occupations that rank low in their work from home measure had larger employment declines (Mongey et al., 2020). Indeed, the risk of becoming unemployed is mitigated by having the option of working from home. If this employment safeguard is more accessible to those who are already economically stable, then one must consider the impacts this may have on a national with high levels of inequality.

Like the work done by Mongey et al., Dingle and Neiman construct their own work from home measure by combing data from the Occupational Information Network and the US Bureau of Labor Statistics to classify the feasibility of remote work. The authors find that 37% of jobs can be done from home and find significant variation across regions and industries. (Dingel & Neiman, 2020). The variation across industries is especially noteworthy since there may be unintended consequences of remote work that could have large economic impacts across these industries.

One such effect, observed by Irlacher and Koch, is that those that work from home receive a wage premium from being able to do so. Using German Qualification and Career Survey Data, the authors conduct a Mincerian regression for workers in Germany performing their job at home and find a 10% wage premium that persists within narrowly defines jobs and workplace controls (Irlacher & Koch, 2020). This wage premium could increase the wage gap and further increase inequality, putting workers that are already less effected by COVID-19 at an even greater advantage.

This aim of this paper is to expand on the work done by Irlacher and Koch by comparing the wage premiums observed in the United States between two industries classified as high and

low work from home capacity: the finance and manufacturing industry, respectively. Following the classification used by Dingle and Neiman, this paper uses the finance and manufacturing industries as proxies for high and low remotely capable jobs[1]. By doing so, I can observe the inequalities between and within the industries regarding access to working remotely as well as the benefits of being able to do so.

3. Data

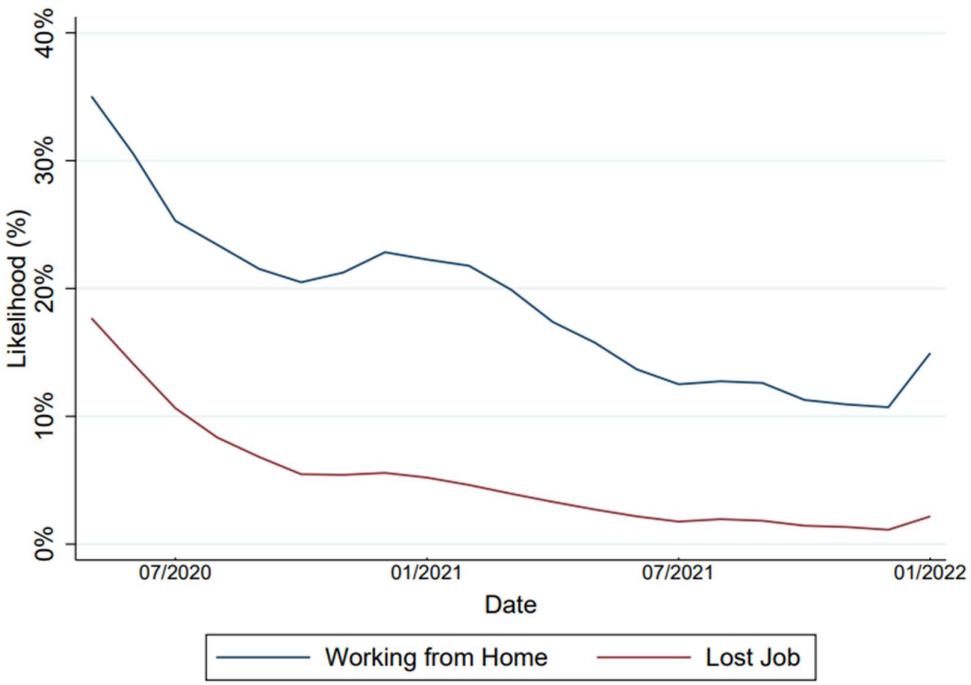
IPUMS Current Population Survey (CPS) is harmonized microdata on the United States labor force and runs monthly dating back from 1962. The United States Census Bureau and the Bureau of Labor Statistics coverers over 65,000 US households and is the source extracted with cross-sectional data from 2010-2022, inclusive of demographic information such as age, sex, race, marital status, and education attainment as well as employment information such as wage, income, class of worker, employment status, industry, and occupation. In May 2020, five questions related to the labor force impacts of COVID-19 were added to the CPS monthly survey. These questions were related to whether the respondent has worked remotely, was unable to work, or was prevented from looking for work due to COVID-19 in the past 4 weeks. A set of summary statistics can be found in Table 1 of the Appendix.

I estimate that 24.7% of CPS respondents worked from home in 2020. That number dropped to 15% in 2021 and remained at 15% in January of 2022, still well above pre-pandemic levels. By focusing on May 2020, the month in which Bick and his co-authors studied, this paper finds estimates consistent with other relevant literature.[2] In comparison, 9% of respondents lost

[1] Dingle and Neiman found that 85% of finance jobs can be done remotely, compared to 36% of manufacturing jobs (Dingle & Neiman, 2020).
[2] Beck et al. find that 35.2% of respondents worked from home in May 2020 (Beck et al., 2020) while I find 35%. Barrero, Bloom, and Davis find that 40% of people aged 20-64 worked from home in May 2020 (Barrero et al., 2020)

their job due to COVID-19 in 2020 while 2.6% and 2.2% lost their jobs in 2021 and 2022, respectively. Figure 1 plots the prevalence of remote work and job loss due to COVID-19 beginning in May of 2020. As indicated in the figure, both working from home and job loss have been on the decline since the onset of the pandemic, with increases in remote work and job loss coinciding with different waves of the pandemic.[3] These results are consistent with the literature, by constricting the data to May 2020, we observe that 35% of people worked from home.

Figure 1: Prevalence of working from home and job loss



Turning attention to inequalities of labor market demographics, Table 2 displays the accessibility of remote work and frequency of job lost in 2020 using a Logit model. The two columns depict the predicted probabilities that the outcome variable, WFH, equals 1[4] for various demographic controls. By fitting a monotonic Logistic CDF link rather than a linear function, we

[3] Increases in the prevalence of breakthrough COVID-19 infections in December 2020 and December 2021 coincide with increases in the percentage of people working from home.

[4] WFH is a Bernoulli variable equaling 1 if the respondent has worked from home due to COVID-19 in the past four weeks

can achieve a more accurate estimate of remote work prevalence. The Logit model is characterized by the equation:

$$Probability(Y_i=1|X_i)=\frac{exp(Xi'\beta)}{1+exp\ (Xi'\beta)}$$

where X_i is a vector of controls[5] and Probability($Y_i|X_i$) represents the conditional mean of the Bernoulli random variable, Y_i. Here Y_i represents the variable WFH and LostJob, which are indicator variables for working from home or becoming unemployed as a result of COVID-19, respectively. Section 6 discusses the robustness of the logit model by comparing the predicted probabilities to that of the Probit model and the Linear Probability Model (LPM).

Regarding gender disparities, in 2020 women were 5.2 percentage points more likely to work from home compared to men. This is consistent with the findings of Reichelt, Makovi, and Sargsyan, who found that women are 7 percentage points more likely to work from home (Reichelt et al., 2020). In terms of job loss, there were no substantial difference between men and women, although women have a slightly higher, yet insignificant chance of losing their job, consistent with results found in the literature.

In accordance with current research, occurrence of remote work increases greatly with level of educational attainment. Figure 2 in the appendix plots the probability of working from home and job loss for varying levels of education. Workers with a college degree are 26 percentage points more likely to be able to work from home than workers with only a high school degree or GED equivalent[6]. The data also indicates a decrease in the likelihood of losing one's job as level of educational attainment increases.

In terms of age, workers in the 18-25 age cohort have a lower work from home probability (19.8% in 2020) than those in the 26-34 cohort, who are the most likely to work from

[5] The controls in the logit regression are sex, education, age bins, income quartiles, and regions.
[6] Barrero et al. find that a unit standard deviation increases in education results in a 6-9 percentage point increase in likelihood of working from home (Barrero et al., 2020)

home (23.3%). This difference is likely because people in the 18-25 cohort are employed in

Table 2: Predicted Probabilities of Working from Home and Job Loss

Standard errors in parentheses

VARIABLES	Predicted Probabilities of WFH	Predicted Probabilities of Job Loss
Less than Highschool	0.0425***	0.0851***
	(0.00379)	(0.00411)
Graduated Highschool	0.0974***	0.0718***
	(0.00235)	(0.00195)
Associate Degree	0.171***	0.0685***
	(0.00427)	(0.00275)
Bachelor's Degree	0.360***	0.0654***
	(0.00379)	(0.00190)
Master's Degree	0.481***	0.0596***
	(0.00600)	(0.00283)
PhD	0.420***	0.0621***
	(0.0125)	(0.00668)
Male	0.185***	0.0671***
	(0.00257)	(0.00148)
Female	0.237***	0.0690***
	(0.00301)	(0.00153)
55-64 Cohort	0.194***	0.0672***
	(0.00373)	(0.00218)
45-54 Cohort	0.200***	0.0739***
	(0.00371)	(0.00228)
35-44 Cohort	0.217***	0.0683***
	(0.00374)	(0.00212)
26-34 Cohort	0.233***	0.0685***
	(0.00418)	(0.00225)
18-25 Cohort	0.198***	0.0568***
	(0.00676)	(0.00264)
Lowest Income Quartile	0.118***	0.132***
	(0.00337)	(0.00346)
2nd Income Quartile	0.175***	0.0758***
	(0.00347)	(0.00223)
3rd Income Quartile	0.233***	0.0553***
	(0.00369)	(0.00184)
Highest Income Quartile	0.320***	0.0445***
	(0.00449)	(0.00172)
Observations	59,130	59,130

*** p<0.01, ** p<0.05, * p<0.1

higher percentages in occupations that have lower remote work feasibility.[7] Indeed, Dey et al. acknowledge that people younger than 25 are more likely to be employed in occupations where work from home is not feasible (Dey et al., 2020). Overall, young people are more likely to be able to perform their job from home, with probability of working from home during the pandemic decreasing slightly with age.[8] There is a small but positive increase in the probability of losing one's job due to COVID-19 as age increases, with a small dip in the 55–64 cohort.

Differences across income quartiles produced staggering results. Figure 3 in the appendix graphs the likelihood of working from home and losing one's job with respect to each income quartile. The data suggests that remote work is more accessible for this at higher income levels. In 2020, 32% of earners in the top income quartile worked from home while only 11.8% of those in the bottom income quartile worked from home. Conversely, those in the top income quartile were much less likely to lose their jobs (4.45% in 2020) compared to those at the bottom quartile (13.2%). For comparison, Li et al. used income quintiles from data in Australia and found that 14.17% of people lost their job in the bottom income quintile while 2.58% lost their jobs in the top quintile (Li et al., 2020).

The IPUMS CPS data confirms the findings of other current research papers and shows that the ability to work from home is unequal in the United States population. Those with high accessibility of remote work more likely to be female, younger, highly educated, and were higher earners prior to the pandemic. These groups are also more like to keep their job throughout the pandemic, especially those that have higher levels of education and earnings. These findings indicate that those that are already vulnerable to the effects of the pandemic are also the ones that

[7] By using finance as a proxy for industries with high remote work accessibility, I estimate that 6.5% of employees are in the 18-25 cohort. The manufacturing industry, which I use as a proxy for low remote work accessibility, has 10.7% of their workforce in the 18-25cohort.

[8] This finding is consistent with that of Brynjolfsson and his co-authors who found that workers below 25 years old were more likely to convert from commuting to work to working from home (Brynjolfsson et al. 2020).

are least likely to benefit from the safeguards of working from home.[9] Next I turn my attention to two industries that function as proxies for high remote work and low remote work capable industries: finance and manufacturing.

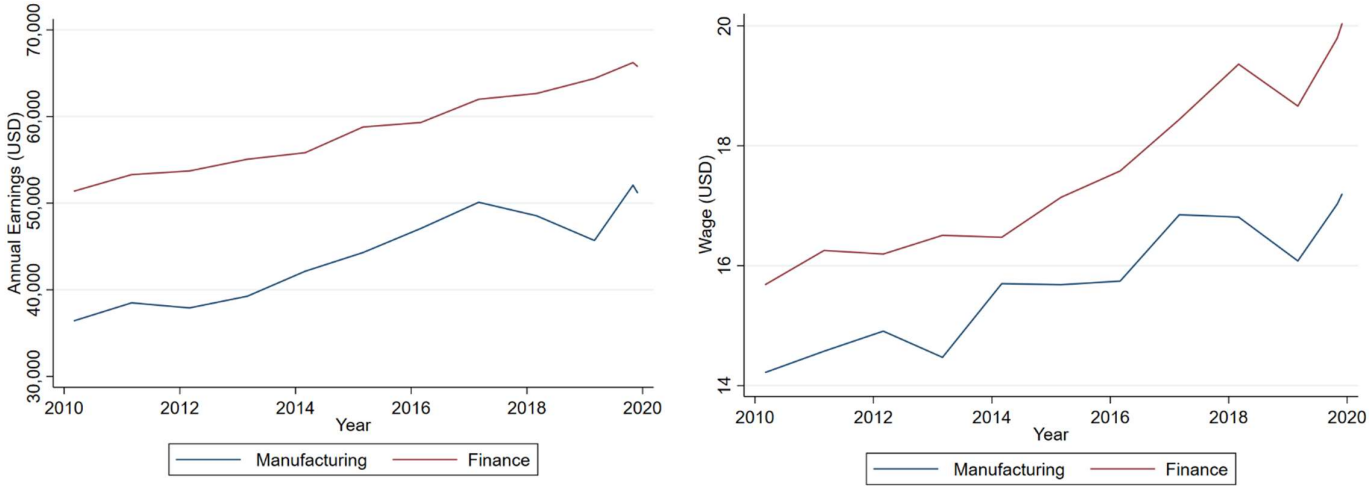
In their work on classifying which jobs are more capable to switching to remote work, Dingle and Neiman compute the share of jobs that can be done from home using the North American Industry Classification System (NAICS). These measures are computed using the Occupational Information Network derived classification of occupations that can be done remotely merged with the BLS's 2018 Occupational Employment Statistics. The authors find significant variation across industries, specifically 85% of occupations in the Finance and Insurance industry has been observed to be able to be done from home with only 36% of occupations in the Manufacturing industry (Dingle and Neiman, 2020).[10] Figure 4 plots the average incomes and wages for the Finance and Manufacturing sector for the years 2010-2019 using the 1990 Census Bureau industrial classification system. In the decade leading up to the pandemic, the average income for the finance industry was 33% higher and wages were 12.2% higher than the average income and wages of the manufacturing industry.[11]

[9] The data indicates a strong negative correlation between being able to work from home and job lost caused by the pandemic. For further details see Table 10 in the Appendix which shows the marginal effects derived from the logit regression.

[10] Both these statistics are when the measures are weighted by wage. The unweighted numbers are 76% and 22% of jobs being able to be done from home for the finance and manufacturing industry respectively.

[11] Average income and wages in the finance industry was $58,905.38 and $17.63 while the same measures for the manufacturing industry were $44,308.50 and $15.71

Figure 4: Wages and Income for the Financial and Manufacturing Industry



Relying on Dingle and Neiman's classification of feasibility to work from home, the finance and manufacturing industries are utilized as proxies for high and low remote work capabilities. The reason for selecting these two industries is that their demographic characteristics are in line with the above findings of typical remote work likeliness. Table 3 depicts the summary statistics for people employed in the finance and manufacturing sector along with the general population over the years 2010-2022. While the finance industry is slightly more represented by women (55% female), the manufacturing industry is predominately male (36% female). The finance industry is more educated on average than the manufacturing industry as well, with 64% of people working in financing having at least a college degree whereas 22% of manufacturing workers have that same educational attainment.[12] Those employed in finance are also wealthier and earn more on average than their manufacturing industry counterparts. The finance industry has the highest concentration of employees residing in the highest income quartile and the lowest concentration in the lowest income quartile. Employees in the manufacturing industry, meanwhile, are mostly concentrated in the middle to lower levels of the

[12] For reference 36% of IPUMS CPS respondents have at least a college degree.

income distribution and are least concentrated in the highest. These findings show that people working in the finance industry are more educated, wealthier, and more likely to be female than the manufacturing industry and the national average.[13] Conversely, those working in manufacturing are less educated, poorer, and more male dominated than that of the national average.

Table 3: Summary Statistics of Finance and Manufacturing

	National Average	**Finance Industry**	**Manufacturing Industry**
Sex			
Male	48.65%	49.98%	64.13%
Female	51.35%	55.02%	35.87%
Education			
No School	0.36%	0.03%	1.10%
Some School	17.81%	2.27%	21.46%
Completed High School	34.01%	21.34%	45.74%
Associate Degree	11.58%	12.69%	9.30%
Bachelor Degree	23.95%	47.91%	17.30%
Masters Degree	10.20%	14.64%	4.72%
Phd	2.09%	1.11%	0.39%
Income Quartiles			
Lowest Quartile	25.52%	12.95%	20.52%
2nd Quartile	24.50%	23.16%	34.95%
3rd Quartile	25.25%	27.88%	26.30%
Highest Quartile	24.73%	36.01%	18.23%
Age Bin			
55-64	23.46%	22.40%	19.73%
45-54	21.17%	24.94%	24.34%
35-44	22.02%	26.33%	24.72%
26-34	18.62%	19.84%	20.53%
18-25	14.74%	6.49%	10.68%

4. Empirical Strategy

In the previous section, the data suggested that workers who are female, have higher levels of education, and more affluent are more likely to work from home and less likely to lose

[13] Where high education, high earning, and being female are all indicators of increased likelihood of working from home in 2020.

their job during the pandemic. Additionally, employees in the finance industry are on average more likely to have higher levels of education, earn more, and be female. For this reason, it would be appropriate to hypothesize that labourers in the financial industry are also more likely to work from home and less likely to go into unemployment than their counterparts in the manufacturing industry.

To test this hypothesis, a Logit regression is run on the predicted probabilities of working from home and losing one's job in 2020 due to COVID-19. Two separate regressions are conducted for whether the respondent was employed in the finance or manufacturing industry to isolate the industry specific effect. The marginal probabilities for being employed in each sector are then retrieved and compared with the national average. Table 4 display the results of this regression. Being employed in finance increases the probability of being able to work from home by 19.5% on average. In terms of job loss, a small but significant decrease in the probability of losing one's job due to the pandemic if a worker is employed in the financial industry is found (1.7% decrease from the national average).

Table 4: Predicted Probabilities of Working from Home and Job Loss in the Finance and Manufacturing Industries

VARIABLES	(1) WFH	(2) Lost Job
Finance	0.195***	-0.0171***
	(0.00242)	(0.00193)
Manufacturing	-0.162***	-2.69e-05
	(0.00879)	(0.00412)
Observations	372,772	673,583

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Conversely, employees in the manufacturing sector have a decreased probability of working from home of 16.2% from the national average[14]. Being employed in manufacturing also has a diminutive and insignificant marginal probability on job loss in 2020, suggesting working in manufacturing had little bearing on being becoming unemployed because of COVID-19. During lockdowns, the government declared a number of manufacturing businesses as essential in order to maintain a functioning economy. An example of such occupations are jobs involving the manufacturing of essential goods such as vaccines. These businesses deemed essential in the manufacturing industry were allowed to stay open and operate as usual, keeping their workers employed but also requiring them to commute to work.[15]

Overall, workers in the finance industry are 35.7% more likely to be able to work from home and 1.7% less likely of becoming unemployed than those working in the manufacturing industry, confirming our hypothesis regarding remote work. In Section 5, the focus is shifted to the effects of being able to work from home have on labor market outcomes such as wage and income and review the empirical strategies implemented.

The main empirical strategy used in this paper is a Mincer type regression of workers who could work remotely during the COVID-19 pandemic as presented in Irlacher & Koch (2020). A stander Mincer earning function is of the form:

$$\ln[w(s,x)]=\alpha_0+\rho s+\beta_0 x+\beta_1 x^2+\varepsilon$$

as described by Jacob Mincer (Mincer, 1958) where s is the number of years of schooling, x is the potential experience of the individual, and ln[w(s,x)] is the logarithm of the wage rate as a function of years of schooling and experience.[16] ρ represents the Mincerian rate of return of an additional year of schooling on the logarithm of the wage rate and ε denotes the regression

[14] In 2020, someone employed in the finance industry was 46.5% likely to work from home, while a manufacturing employee was only 11.1% likely.

[15] While be allowed to continue to get into work is economically good, there could also be negative health consequences of doing so during a pandemic.

[16] Here s is measured by the variable educ99 which indicated the years of educational attainment by the individual, x is represented by the variable exp=age-educ99-6 (measured in levels and squared)

residual. This paper assumes that the returns to education are the same for all levels of schooling and that the regression residual is mean zero ($E(\varepsilon|s, x)=0$). Assuming mean independence of the regression residual allows our estimates of returns to education to be unbiased. While the model does not consider aspects such as age-varying returns to education and cost of schooling, we are still able to gain insight into the impact on wages of attaining an additional year of education.

Additionally, the Mincer model does not consider that people who have higher levels of education are more likely to be employed.[17] However, by using potential experience as a proxy for actual labor market experience, this issue is rectified by allowing the experience variable to not be affected by employment status.

To identify the labor market returns of working from home, the empirical strategy used in Irlacher and Koch is employed to expand upon the Mincer regression by including the dummy variable for a worker being able to work from home in the year 2020 as well as a full set of controls and fixed effects. To account for heterogeneity in wages on an individual basis, individual demographic controls for age and indicator variables for female, married, and native born in the United States are included in the model. Industry and geographical region fixed effects are also incorporated to further control for unidentified wage heterogeneity. Doing so allows for identification of a wage premium associated with being able to work from home during the pandemic. See the Results section for discussion on the estimation of this regression.

Combing the approach of Irlacher and Koch with the remote work feasibility classification of Dingle and Neiman, a separate wage premium is identified for the finance and manufacturing industry. This is achieved by running the same Mincer type regression as before, but isolating the population that works in finance, then those that work in manufacturing using the full set of controls. Coupling this with the realization of work from home accessibility in both industries identified in the previous section, this paper identifies how the disparities between and

[17] Indeed, in the data I find a positive correlation between years of education and probability of being employed.

within these two industries may become exacerbated by COVID-19 as well as the persistency of these inequalities even as the pandemic comes to an end.

5. Results & Discussion

The results of the standard OLS Mincer regression without additional controls are shown on Table 5. The returns on education are 6.5%. The result remains significant at the 1%, 5% and 10% level. This suggests that education a good investment on future earnings. Typical OLS estimates in the United States from a Mincer earning function are about 8%. The estimates are consistent with typical findings, therefore implying that a Mincer regression will give valid results if expanded. [18]

Table 5: Mincer Regression Results

VARIABLES	logwage
Education	0.0653***
	(0.000338)
Experience	0.00693***
	(6.01e-05)
Experience Sq	-8.41e-07***
	(7.70e-08)
Observations	246,724
R-squared	0.161

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

[18] Figure 5 in the appendix plots the correlation between years of education and average wages.

Table 6: Wage Premium

	Dependent Variable: Logarithm of Hourly Wage			
2020	(1)	(2)	(3)	(4)
WFH	0.260***	0.157***	0.150***	0.120***
	(0.00643)	(0.00597)	(0.00594)	(0.00587)
Observations	47,714	47,714	47,714	47,714
R-squared	0.033	0.224	0.234	0.345
2021	(1)	(2)	(3)	(4)
WFH	0.281***	0.176***	0.168***	0.125***
	(0.00647)	(0.00598)	(0.00595)	(0.00578)
Observations	74,993	74,993	74,993	74,993
R-squared	0.024	0.204	0.214	0.323
2022	(1)	(2)	(3)	(4)
WFH	0.274***	0.170***	0.163***	0.131***
	(0.0219)	(0.0205)	(0.0204)	(0.0203)
Observations	6,037	6,037	6,037	6,021
R-squared	0.025	0.187	0.200	0.311
Demographic Controls:	No	Yes	Yes	Yes
Region FE:	No	No	Yes	Yes
Industry FE:	No	No	No	Yes

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table 6 depicts the results of regressing the logarithm of hourly wage on the indicator variable for working from home in the past four weeks because of the COVID-19 pandemic. In the first year of the pandemic (2020), a 12% wage premium is found for those that were able to work from home, after controlling for worker demographics, geographical region fixed effects, and industry fixed effects.[19] This premium is significant even at the 1% level. The wage premium

[19] The controls for this regression are education attainment, age, experience, experience squared, and indicator variables for female, marital status, and native to the United States as well as industry and region fixed effects.

remains even after controlling for education, indicating that the work from home wage premium is not solely due to differences in education levels.

The mechanisms driving this wage premium are topics of debate in the literature. Classical Economic Theory tells us that workers' wages are tied to their marginal product. Therefore, if workers are more productive when they are working from home, then their wages would increase to reflect that.[20] This idea has been proposed prior to the pandemic. A 2015 randomized control trial of a Chinese travel agency found that workers who were allowed to work from home saw their productivity increase by 13% (Bloom et al., 2015).[21] The authors found that wages of employees who could work remotely rose 9.9%. The drivers of this productivity boost may be due to higher employee morale, less time wasting due to office distractions, or decreases in commuting times. Irlacher and Koch argue that this is the mechanism driving the work from home wage premium. Productivity can be boosted by allowing employees to work from home, as seen by the respondents of the May 2020 Survey of Working Arrangements and Attitudes, which reported to having better than expected work from home experiences and increased productivity during the pandemic (Barrero et al., 2020).[22] However, unlike the Chinese travel agency, where employees' wages are tied to their performance, many full-time employees in the United States have sticky wages that are fixed for periods of time. Therefore, the wage premium found in this paper is unlikely to be driven by boosts in productivity. Instead, the wage premium of working from home is more likely to be a result of higher paying industries being able to transition to remote work more easily. From 2019 through January of 2022, wages of employees in the financial industry were 13.9% higher than those employed in manufacturing. At the same time, employees in the finance industry are 35.7% more

[20] Classical theory tells us that in a competitive market, workers are paid their marginal product. As such, if workers' productivity increases, their marginal productivity increases, therefore their wages must increase.
[21] The treatment group of this experiment are employees of the travel agency that were allowed to work from home and the control group are employees that remained in the office.
[22] The authors predicted a 4.6% increase in productivity being driven by reduced commuting times.

likely to work from home. Since the population of employees working from home have a higher concentration of people working in high paying industries such as finance, it is reasonable that wages will be higher than the population that is unable to work from home.[23] Furthermore, by calculating wage growth for both industries over the two-year period from 2019-2021, this paper found that finance wages grew 8.6% while manufacturing wages grew 10.3%. If the wage premium was being driven by increases in productivity, we would expect to see more wage growth in industries with higher remote work accessibility.

This paper also estimates that the wage premium increased in 2021 to 12.5% and again further increased for the first month of 2022 to 13.1%. This suggests that the wage premium increased over the past 3 years due to low paid workers returning to in person work, while higher paid workers remain remote. As the pandemic winds down and businesses begin to reopen, it will be lower ability, lower paid workers that return to in person work. Industries like Finance that were high work from home feasibility to begin with are more likely to remain remote even as lockdown restrictions begin to ease.[24] This evidence further suggests that the wage premium is due to a higher fraction of remote workers being employed in higher paying industries.

[23] Figure 6 in the appendix plots the average wage of those that work from home versus those that do not. The figure suggests that employees who work from home earn significantly higher wages on average than those that do not.

[24] Barrero et al. estimate that 20% of workdays will be supplied from home after the pandemic ends (Barrero et al., 2020)

Table 7: Wage Premium for Finance and Manufacturing Industry

VARIABLES	(1) Finance 2020	(2) Finance 2021	(3) Finance 2022
WFH	0.142***	0.109***	0.131**
	(0.0175)	(0.0156)	(0.0548)
Constant	169.0***	6.119***	2.311***
	(25.83)	(1.772)	(0.160)
Observations	2,036	3,059	251
R-squared	0.195	0.130	0.261
VARIABLES	(1) Manufacturing 2020	(2) Manufacturing 2021	(3) Manufacturing 2022
WFH	0.205***	0.0640	-0.0805
	(0.0692)	(0.0598)	(0.165)
Constant	262.8***	179.6***	2.654***
	(73.08)	(34.05)	(0.222)
Observations	835	1,258	95
R-squared	0.237	0.175	0.388

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

This paper has thus far presented evidence that higher paying industries have higher ability to work from home, and that the employees of those industries generally fair better during the pandemic. The question remains about how these industries differ in remote work impacts. Table 7 shows the results of the Mincer regressions characterized in the previous section for employees in the finance and manufacturing industries separately. In 2020, the finance industry had a 14.2% wage premium, over 18% higher than the national population. In 2020, 46.5% of those employed in finance worked from home. Earlier estimates found that those who are more educated and earning a higher wage are more likely to work from home. The higher wage premium implies those that those working from home in 2020 in the finance industry are in a

higher pay bracket than that of the 53.5% that could not work from home. Dingle and Neiman argued that people employed in higher paying occupations such as managers are more likely to work from home that production workers (Dingle and Neiman, 2020.) If we accept this argument, then the estimates indicate that within industries those that are paid more are the ones that work from home more, resulting in a perceived wage premium. Meanwhile, the manufacturing industry experienced a 20.5% wage premium for being able to work from home. Recall that the manufacturing industry generally had less access to remote work, with only 11.1% of employees in the industry working from home in 2020. As with the finance industry, the previous estimate suggests that of the 11.1% working from home in the manufacturing industry, it is most likely to be the higher paid roles that have that option.[25] Therefore, the small fraction of manufacturing workers that can work from home are more likely to be at the top of the wage and income distribution, resulting in a higher wage premium for that industry. The results for 2020 in Table 7 bolster the argument that the work from home wage premium is not a result of the boosted productivity, but rather the higher accessibility of remote work by not only high ability industries but also high ability employees.[26]

The persistence of working from home and the wage premium is evident by the results of Table 7. From 2020 to 2021, the rate of remote work decreased 30% for the finance industry and 37% for the manufacturing industry. As the economy and businesses began to reopen, manufacturing employees went back to in person work at higher rates than finance employees. Meanwhile the wage premium remained significant and above 10% for finance, while becoming insignificant and reducing substantially in magnitude for manufacturing. [27] The work done by Barrero and his co-authors reflect this theory, estimating that 20% of workdays will be supplied

[25] Krantz-Kentkrantz found that states with higher concentrations of management, professional and related occupations were more likely to work from home. (Krantz-Kentkrantz, 2019).
[26] Here I rely on the assumption that high ability is associated with high pay. This argument may be up for debate regarding the definition of "ability." For simplicities sake, I refer to high ability employees as employees in the upper tier of the wage and income distribution.
[27] In 2022, the wage premium for manufacturing becomes negative, suggesting a disbenefit for remote work in that industry.

from home even after the pandemic and that the benefits of remote work will mainly go to the highly educated and highly paid (Barrero et al., 2020). As industries that are made up of lower educated and lower paid employees return to work at higher rates than their better paid and educated counterparts, the benefits are remote work further benefit those at less economic risk.[28]

6. Robustness Check

In this section, the findings are verified by conducting a series of robustness checks. I begin by comparing the results of the Logit regression implemented in Section 3 with the estimates of the Linear Probability Model (LPM) and a Probit Model using the same controls. The LPM model uses a linear function to estimate the effects of the independent variables on the probability of working from home. This model does not follow the rule of probabilities when the dependent variable is binary.[29] The Probit model uses a Gaussian CDF as its link function to model the conditional probability of working from home (as compared to the Logistic CDF being used in the Logit Model). My choice of the Logit Model is mostly due to computational ease, as the logistic CDF is mathematically simpler, although either model would yield valid results. Table 8 reports the result of the three regressions.[30] The Logit, Probit, and Linear Probability models all give significant and similar marginal results for each of the worker demographics studied. The Probit Model gives a larger difference between male and female work from home probabilities, more in line with the findings of Reichelt et al. Overall the Logit and the Probit estimates track each other nicely while the LPM model has some insignificance regarding the education marginalities, but overall, not to difference from the other two results. The results of

[28] This does not consider the health benefits of remote work during a pandemic, which is outside the scope of this paper. For more information on the health impacts resulting from remote work, see Ozguzel et al. 2020.
[29] The Linear Probability model is not bounded by 1 and 0. The regression could in theory estimate a probability of working from home as being greater than 1 or less than 0, which would tarnish my results.
[30] The Logit and Probit results are the marginal effects on the probability that WFH=1. The LPM regression is a standard OLS regression with the indicator variable WFH as the dependent variable. The controls for all three regressions are sex, education, age, geographic region, and income quartile.

the robustness check verify that the estimates found in Section 3 are valid and the logit regression is a reasonable one, so we can accept the findings with confidence.

Table 8: Robustness Check

VARIABLES	(1) Logit	(2) Probit	(3) LPM
Female	0.0523***	0.0612***	0.0534***
	(0.00359)	(0.00376)	(0.00341)
45-54	0.00610	0.00659	0.00735
	(0.00494)	(0.00522)	(0.00489)
35-44	0.0232***	0.0236***	0.0253***
	(0.00494)	(0.00522)	(0.00482)
26-34	0.0385***	0.0396***	0.0408***
	(0.00532)	(0.00557)	(0.00503)
18-25	0.00349	0.000910	0.0204***
	(0.00773)	(0.00781)	(0.00645)
Didn't Graduate High School	0.0127	0.0112	0.0110
	(0.0211)	(0.0232)	(0.0423)
High School Graduate	0.0676***	0.0644***	0.0400
	(0.0209)	(0.0229)	(0.0419)
Associates	0.141***	0.137***	0.100**
	(0.0212)	(0.0232)	(0.0421)
Bachelor's	0.330***	0.327***	0.294***
	(0.0211)	(0.0232)	(0.0420)
Master's	0.451***	0.449***	0.423***
	(0.0216)	(0.0236)	(0.0422)
PhD	0.390***	0.390***	0.366***
	(0.0242)	(0.0261)	(0.0431)
2nd Income Quartile	0.0569***	0.0577***	0.0464***
	(0.00475)	(0.00492)	(0.00496)
3rd Income Quatile	0.114***	0.118***	0.104***
	(0.00496)	(0.00515)	(0.00518)
Top Income Quartile	0.202***	0.210***	0.202***
	(0.00568)	(0.00582)	(0.00559)
Observations	59,130	59,130	59,130

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

By running a Mincer regression like the one used by Irlacher and Koch, a 12% wage premium is found from being able to work from home. To see if the wage premium is valid, this paper moves away from the Mincer formulation and run a regression like that of Mas and Pallais. Their work in 2017, prior to the COVDI-19 pandemic, assessed employee interest in remote

work. They estimate that workers with formal work from home arrangements earn about 10% (in terms of their weekly wage) more than those that don't (Mas and Pallais, 2017). Following their formulation, we run regression of working from home on the log of hourly wage, with controls for hours worked, age, geographical region, education, race, marital status, indicator variables for whether the individual is part time and if the individual was born outside of the United States, and industry fixed effects.

Table 9 displays the result of these separate regressions. The Mincer formulation estimated wage premiums of 12% nationwide, 14.2% for finance and 20.5% for Manufacturing. The Mas and Pallais construction give results of 12.2% nationwide, 14.2% for finance, and 25.3% for manufacturing, with all 3 results being significant at the 1% level. These results verify the estimates found in Section 5 and even bolster the argument that those that can work from home are more likely to be high ability workers.[31]

Table 9: Wage Premium Robustness Check

VARIABLES	(1) Total	(2) Finance	(3) Manufacturing
WFH	0.122***	0.142***	0.253***
	(0.00591)	(0.0175)	(0.0706)
Constant	2.236***	2.151***	2.375***
	(0.0164)	(0.0826)	(0.101)
Observations	44,884	1,967	787
R-squared	0.358	0.192	0.281

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

[31] The 25.3% wage premium for the manufacturing industry signals even further that within that sector, it is the top earners that are more likely to work from home.

7. Conclusion

The COVID-19 pandemic has ushered in unprecedented changes in the labor markets, particularly regarding remote working environments. There has been drastic normalization and implementation of remote work all over the world, across all industries and levels of which have never been seen before. To examine the effects of this shift in remote work, accessibility is compared to worker demographics relevant to their respective industry. Through the exploration of these and understanding the benefits, researchers can further anticipate how the labor market will operate moving forward.

By using IPUMS CPS data of the United States population, 24.7% of respondents worked from home in 2020 because of the pandemic, surpassing all historical figures. This number decreased to 15% in 2021, although still higher than pre-pandemic levels, implying persistence of remote work. These figures correlate with the rate of job loss caused by the pandemic, with 9% of CPS respondents losing their job in 2020 due to COVID-19, 2.6% in 2021, and 2.2 in January of 2022.

Through a Logit regression, while controlling for sex, age, education, geographical region, and income levels, we conclude that remote work is mostly accessible to those that are more educated, female, younger, and on the higher end of the income distribution. This signals the inequality of being able to work from home and how those that were already shielded from the negative economic consequences of the pandemic are also those that are most likely to benefit from accessibility of alternative work arrangements.

Narrowing the focus to two specific industries acting as proxies for high and low remote work feasibility, we find that the finance industry pays their employees 13.9% more on average than the manufacturing industry. Employees of the finance industry on average are also more educated, placed higher in the income distribution, and female. It was therefore unsurprising to find that employees in the financial industry were 35.7% more likely to work from home in 2020.

This suggests that higher paying industries are the ones that can work from home.

Additionally, there was a 12% wage premium from being able to work from home during the first year in the pandemic. The premium was 14.2% for the financial industry and 20.5% for the manufacturing industry, suggesting that within these industries, those higher paid workers have more access to remote work. The persistence of the wage premium for the finance industry and the disappearance of the premium for the manufacturing industry further indicates that only industries with high wages will continue to benefit from remote work as the pandemic comes to an end.

Overall, the results suggest that those that are economically stable are less effected from the labor market disruptions caused by COVID-19. As lockdowns restrictions ease and business goes back to normal, these same individuals who were shielded from the negative impacts of the pandemic will emerge better off and may remain so in the future.

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9. Appendix

Table 1: Summary Statistics of IPUMS CPS Respondents

Variable	Description	2019 Mean	2020 Mean	2021 Mean	2022 Mean
WFH	=1 if worked from home due to COVID-19 in the past 4 weeks	.	.247	.152	.15
lostjob	=1 if lost job due to COVID-19 in the past 4 weeks	.	.09	.026	.022
age	Age	39.26	41.14	41.05	41.12
exp	Potential Experience	24.02	25.52	25.44	25.49
hourwage	Hourly wage	18.79	19.75	20.23	20.75
annualearn	Annual income	52,776.41	55,611.33	56,797.88	58,080.57
female	=1 if female	.51	.51	.51	.51
cit	=1 if a citizen of the USA	.94	.95	.95	.94
native	=1 if born in the USA	.86	.88	.87	.87
child	=1 if has a child at home	.29	.28	.28	.28
emp	=1 if currently employed	.48	.45	.46	.47
unemp	=1 if currently unemployed	.031	.056	.042	.043
fulltime	=1 if employed fulltime	.358	.325	.344	.339
parttime	=1 if employed parttime	.106	.109	.102	.104
selfemployed	=1 if self-employed	.051	.054	.054	.053
nohs	=1 if didn't graduate highschool	.116	.102	.103	.103
hs	=1 if only graduated highschool	.221	.229	.232	.234
college	=1 if graduated college	.161	.171	.171	.173
master	=1 if attained masters degree	.069	.075	.075	.075
phd	=1 if received Phd	.014	.015	.016	.016
associate	=1 if attained associates degree	.077	.082	.08	.079
finance	=1 if works in finance sector	.032	.032	.032	.032
manf	=1 if works in manufacturing	.007	.006	.006	.006

Figure 2: Working from Home and Job Loss Prevalence by Education Level

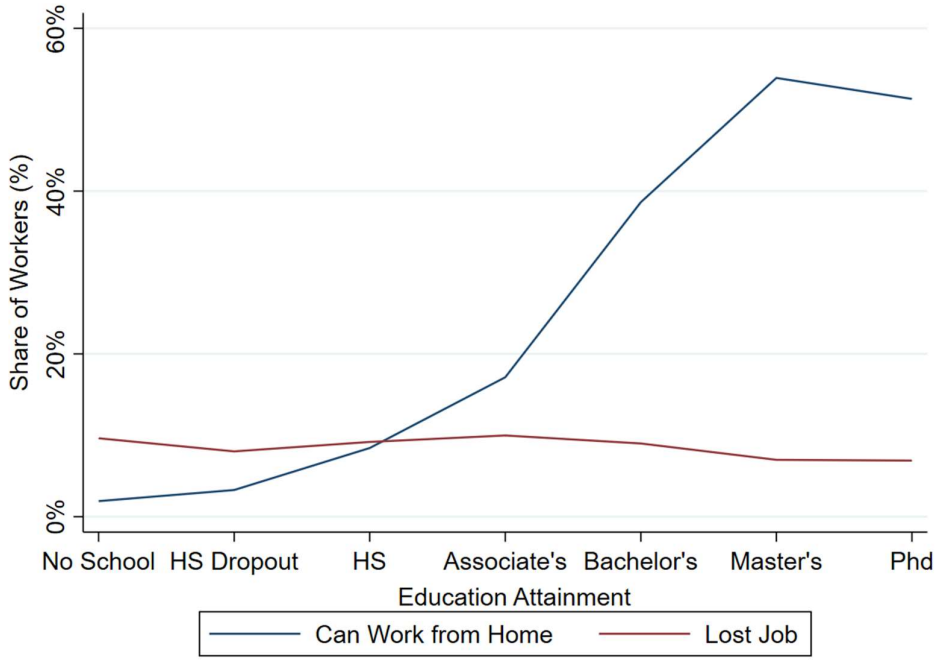

Figure 3: Working from Home and Job Loss Prevalence by Income Quartile

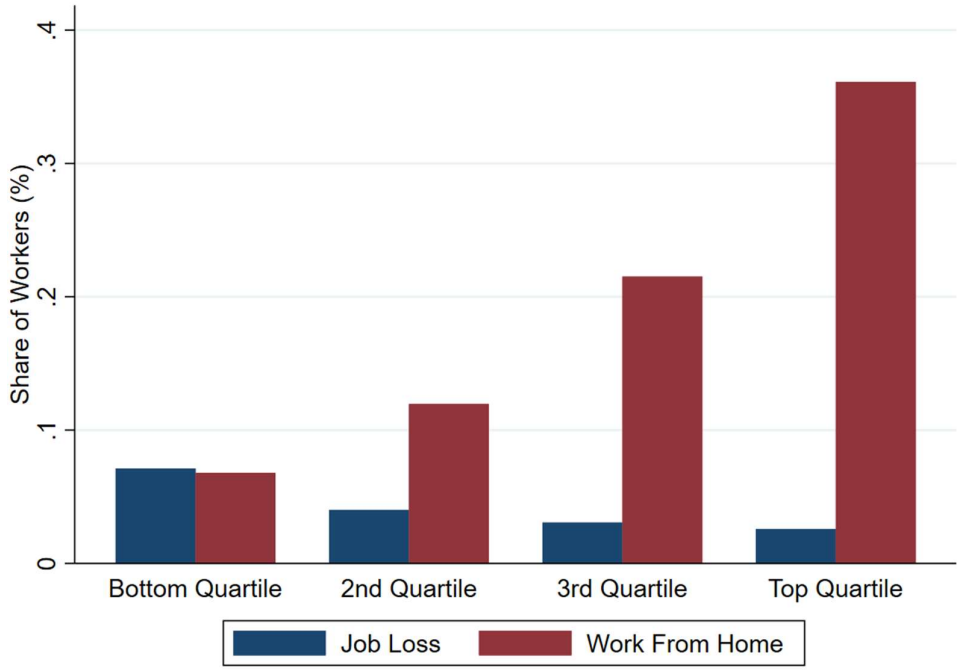

Table 10: Marginal Probabilities of Working from Home and Job Loss

VARIABLES	(1) WFH	(1) Job Loss
Female	0.0523***	0.00188
	(0.00359)	(0.00209)
45-54	0.00610	0.00675**
	(0.00494)	(0.00312)
35-44	0.0232***	0.00114
	(0.00494)	(0.00301)
26-34	0.0385***	0.00133
	(0.00532)	(0.00310)
18-25	0.00349	-0.0104***
	(0.00773)	(0.00333)
Didn't Graduate High School	0.0127	-0.0161
	(0.0211)	(0.0271)
High School Graduate	0.0676***	-0.0294
	(0.0209)	(0.0269)
Associates	0.141***	-0.0327
	(0.0212)	(0.0270)
Bachelor's	0.330***	-0.0358
	(0.0211)	(0.0269)
Master's	0.451***	-0.0416
	(0.0216)	(0.0270)
PhD	0.390***	-0.0391
	(0.0242)	(0.0277)
2nd Income Quartile	0.0569***	-0.0564***
	(0.00475)	(0.00391)
3rd Income Quatile	0.114***	-0.0769***
	(0.00496)	(0.00398)
Top Income Quartile	0.202***	-0.0877***
	(0.00568)	(0.00409)
Observations	59,130	59,130

Figure 5: Wages by Years of Education

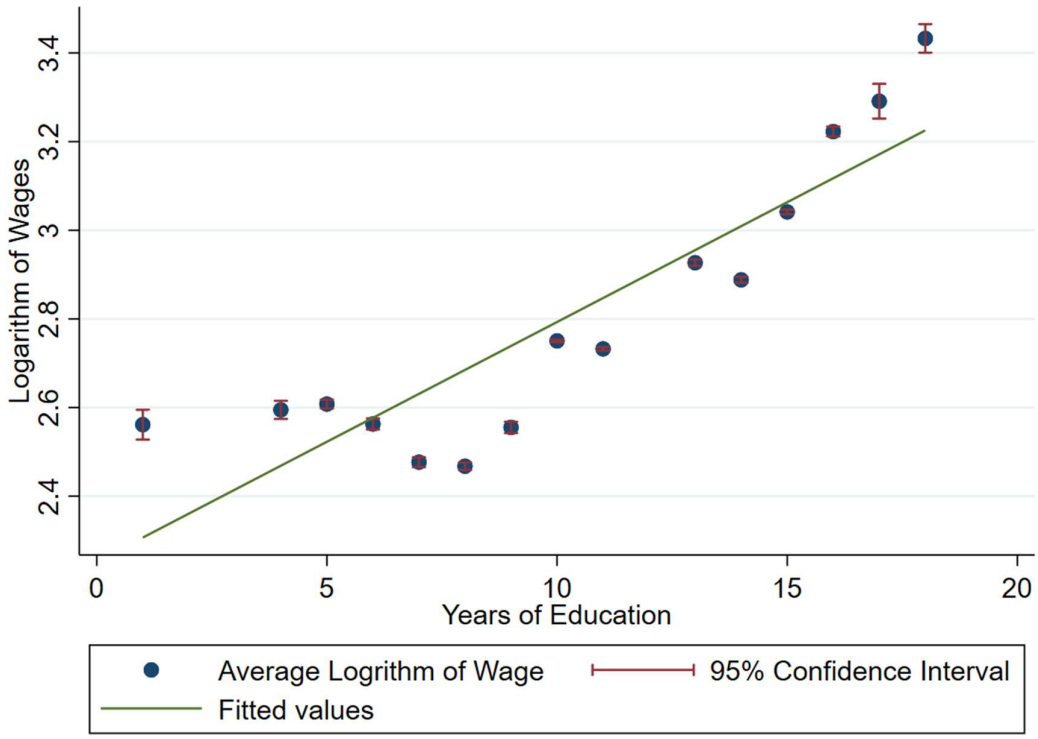
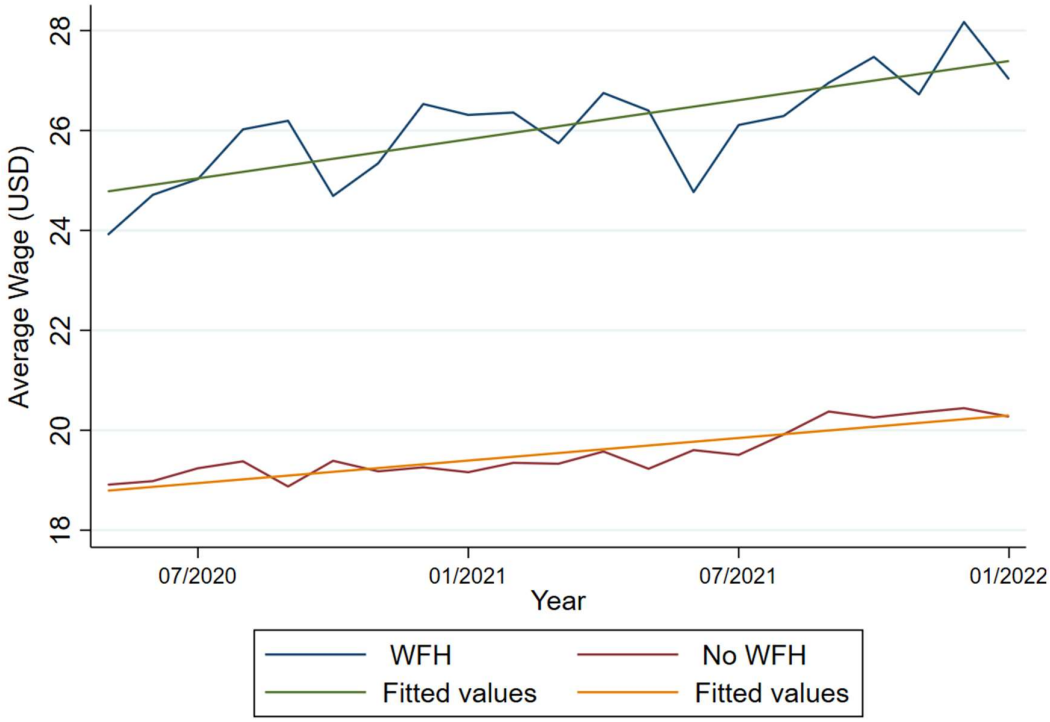
Figure 6: Wages of those that can Work from Home versus those that cannot